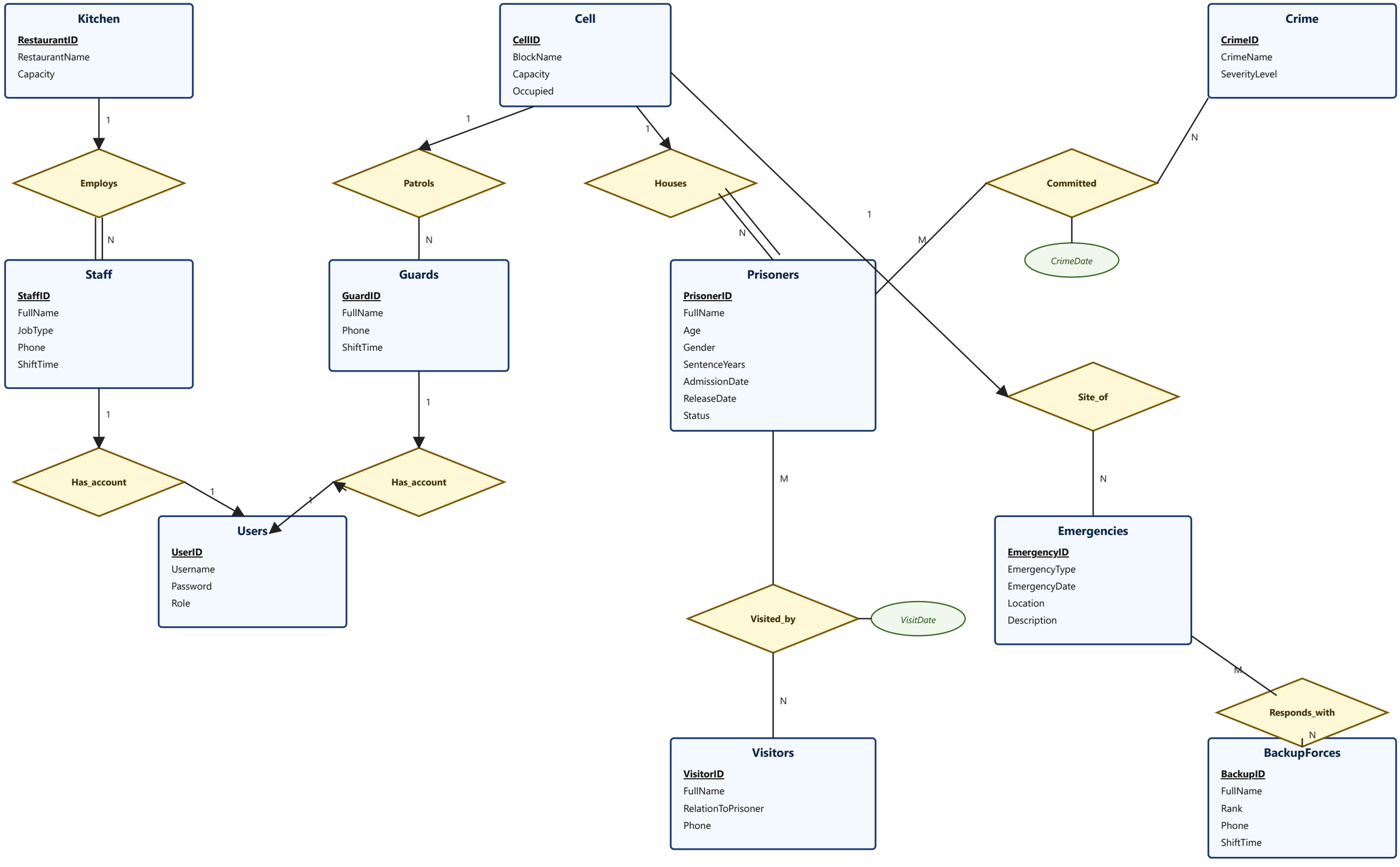
Prison Management System

Entity–Relationship Diagram & Relational Schema — Chen / Silberschatz notation (Ch 7)

Entity set (rectangle) Relationship set (diamond) Descriptive attribute (oval) — line = many (partial participation) → arrow = one = double line = total participation

attr = primary key *attr* = foreign key

1. ER Diagram (Chen-style)


Kitchen
RestaurantID
RestaurantName
Capacity
Employs
Staff
StaffID
FullName
JobType
Phone
ShiftTime
Has_account
Users
UserID
Username
Password
Role
Cell
CellID
BlockName
Capacity
Occupied
Patrols
Guards
GuardID
FullName
Phone
ShiftTime
Has_account
Houses
Prisoners
PrisonerID
FullName
Age
Gender
SentenceYears
AdmissionDate
ReleaseDate
Status
Visited_by
VisitDate
Visitors
VisitorID
FullName
RelationToPrisoner
Phone
Crime
CrimeID
CrimeName
SeverityLevel
Committed
CrimeDate
Site_of
Emergencies
EmergencyID
EmergencyType
EmergencyDate
Location
Description
Responds_with
BackupForces
BackupID
FullName
Rank
Phone
ShiftTime
1
N
M

Notation follows Silberschatz Ch 7. **Arrow (→)** at an end = "one". **Plain line** = "many". **Double line** = total participation (every entity must participate). Primary keys are underlined inside each rectangle. Descriptive attributes of a relationship (e.g. *CrimeDate*, *VisitDate*) hang off the diamond in an oval.

2. Entity Sets & Attributes

Entity Set	Primary Key	Other Attributes	Domain Notes
Cell	CellID	BlockName, Capacity, Occupied	Capacity, Occupied ≥ 0; Occupied ≤ Capacity (semantic)
Crime	CrimeID	CrimeName, SeverityLevel	SeverityLevel ∈ {Low, Medium, High, Critical}
Kitchen	RestaurantID	RestaurantName, Capacity	
Staff	StaffID	FullName, JobType, Phone, ShiftTime	
Guards	GuardID	FullName, Phone, ShiftTime	
Prisoners	PrisonerID	FullName, Age, Gender, SentenceYears, AdmissionDate, ReleaseDate, Status	derived: *YearsServed* = (Now() − AdmissionDate)
Visitors	VisitorID	FullName, RelationToPrisoner, Phone	
Emergencies	EmergencyID	EmergencyType, EmergencyDate, Location, Description	EmergencyDate defaults to Now()
BackupForces	BackupID	FullName, Rank, Phone, ShiftTime	
Users	UserID	Username (unique), Password, Role	Role ∈ {Admin, Staff, Guard}

3. Relationship Sets — Cardinality & Participation Rules

Relationship	Between	Cardinality	Participation	Descriptive Attributes	Business Rule
Employs	Kitchen — Staff	1 : N	Staff: **total**; Kitchen: partial	—	Every staff member works in exactly one kitchen.
Patrols	Cell — Guards	1 : N	Guards: partial; Cell: partial	—	A guard may be assigned to at most one cell at a time.
Houses	Cell — Prisoners	1 : N	Prisoners: **total**; Cell: partial	—	Every prisoner is housed in exactly one cell.
Site_of	Cell — Emergencies	1 : N	Both partial	—	An emergency may occur in a cell, or in a common area (NULL CellID).
Committed	Prisoners — Crime	M : N	Prisoners: **total**; Crime: partial	CrimeDate	A prisoner is convicted of one or more crimes; a crime type may apply to many prisoners.
Visited_by	Prisoners — Visitors	M : N	Both partial	VisitDate	Visit log: same visitor may visit same prisoner many times on different dates.
Responds_with	Emergencies — BackupForces	M : N	Both partial	—	Multiple backup units may respond to one emergency; one unit may respond to many.
Has_account (Staff)	Staff — Users	1 : 1	Both partial	—	A staff member may have a system login; a user account may belong to a staff member.
Has_account (Guards)	Guards — Users	1 : 1	Both partial	—	A guard may have a system login; a user account may belong to a guard.

4. Relational Schema (Reduction to Tables — Ch 7_3)

Primary keys are **underlined**. Foreign keys are *italic-orange*. Attributes that are both PK and FK are underlined-italic-orange.

Cell(<u>CellID</u>, BlockName, Capacity, Occupied)

Crime(<u>CrimeID</u>, CrimeName, SeverityLevel)

Kitchen(<u>RestaurantID</u>, RestaurantName, Capacity)

Staff(<u>StaffID</u>, FullName, JobType, Phone, ShiftTime, RestaurantID) // FK: RestaurantID → Kitchen(RestaurantID)

Guards(<u>GuardID</u>, FullName, Phone, ShiftTime, CellID) // FK: CellID → Cell(CellID)

Prisoners(<u>PrisonerID</u>, FullName, Age, Gender, SentenceYears, AdmissionDate, ReleaseDate, CellID, Status) // FK: CellID → Cell(CellID)

Visitors(<u>VisitorID</u>, FullName, RelationToPrisoner, Phone)

BackupForces(<u>BackupID</u>, FullName, Rank, Phone, ShiftTime)

Emergencies(<u>EmergencyID</u>, EmergencyType, EmergencyDate, Location, Description, CellID) // FK: CellID → Cell(CellID), nullable

PrisonerCrime(<u>PrisonerCrimeID</u>, PrisonerID, CrimeID, CrimeDate) // FKs: PrisonerID → Prisoners, CrimeID → Crime · alt. PK: (PrisonerID, CrimeID, CrimeDate)

Visits(<u>VisitID</u>, PrisonerID, VisitorID, VisitDate) // FKs: PrisonerID → Prisoners, VisitorID → Visitors

EmergencyBackup(<u>EmergencyBackupID</u>, EmergencyID, BackupID) // FKs: EmergencyID → Emergencies, BackupID → BackupForces

Users(<u>UserID</u>, Username, Password, Role, StaffID, GuardID) // FKs: StaffID → Staff (nullable), GuardID → Guards (nullable); Username UNIQUE; CHECK exactly one of (StaffID, GuardID) NOT NULL when Role ≠ 'Admin'

5. Reduction Rules Applied (mapping ER → Relational, per Ch 7_3)

1. **Strong entity sets → one table each.** Cell, Crime, Kitchen, Staff, Guards, Prisoners, Visitors, BackupForces, Emergencies, Users are all strong — each becomes a relation with the same attributes and the same primary key. (Ch 7_3 — "Strong Entity-Sets with simple attributes")
2. **1 : N relationships → foreign key on the "many" side.** No separate table is needed for the relationship; instead, the primary key of the "one" side is added as a foreign key column to the "many" side. Applied to:
   - *Employs* → `RestaurantID` added to **Staff**
   - *Patrols* → `CellID` added to **Guards**
   - *Houses* → `CellID` added to **Prisoners**
   - *Site_of* → `CellID` added to **Emergencies** (nullable because participation is partial)
3. **M : N relationships → new junction table.** The relationship becomes its own relation whose attributes are the union of the participating entities' primary keys plus any descriptive attributes of the relationship. Applied to:
   - *Committed* → **PrisonerCrime**(PrisonerID, CrimeID, CrimeDate)
   - *Visited_by* → **Visits**(PrisonerID, VisitorID, VisitDate)
   - *Responds_with* → **EmergencyBackup**(EmergencyID, BackupID)

   A surrogate PK (`PrisonerCrimeID` etc.) is used in the built database for convenience; the natural composite PK of the relationship would otherwise be the union of the two foreign keys.
4. **1 : 1 relationships → foreign key on either side (chosen by participation).** The *Has_account* relationships between Users and (Staff / Guards) become two nullable foreign keys (`StaffID`, `GuardID`) on the **Users** side, because participation on the Staff/Guards side is partial (not every staff member has a login). (Ch 7_3 — "Schema Combination — One-to-one")
5. **Descriptive attributes of a relationship.** *CrimeDate* on *Committed* and *VisitDate* on *Visited_by* are stored as columns of the corresponding junction tables, exactly as the textbook prescribes for an attribute hanging off a diamond. (Ch 7_3 — "Relationship-Set Example: borrower")
6. **Domain constraints → CHECK / validation rules.** `Crime.SeverityLevel ∈ {Low, Medium, High, Critical}` and `Users.Role ∈ {Admin, Staff, Guard}` are implemented as validation rules in the resulting database. (Ch 7_1 — "Constraints")
7. **Referential integrity.** Every foreign key is declared with referential integrity enforced + cascade update, matching the constraint type defined in the ER model. (Ch 7_1 — "Keys")
8. **No weak entity sets in this schema.** Every entity is independently identifiable by its own surrogate ID, so the weak-entity rules from Ch 7_1 (dashed-underline discriminator, double-diamond identifying relationship, composite PK = owner PK ∪ discriminator) are not applied here. They would apply if, for example, *Visit* had no surrogate VisitID and were identified only by (PrisonerID, VisitorID, VisitDate).

6. SQL DDL Summary (matches the built .accdb)

```
Cell            (CellID, BlockName, Capacity, Occupied)
Crime           (CrimeID, CrimeName, SeverityLevel)
Kitchen         (RestaurantID, RestaurantName, Capacity)
Staff           (StaffID, FullName, JobType, Phone, ShiftTime, RestaurantID→Kitchen)
Guards          (GuardID, FullName, Phone, ShiftTime, CellID→Cell)
Prisoners       (PrisonerID, FullName, Age, Gender, SentenceYears, AdmissionDate, ReleaseDate, CellID→Cell, Status)
Visitors        (VisitorID, FullName, RelationToPrisoner, Phone)
BackupForces    (BackupID, FullName, Rank, Phone, ShiftTime)
Emergencies     (EmergencyID, EmergencyType, EmergencyDate, Location, Description, CellID→Cell)
PrisonerCrime   (PrisonerCrimeID, PrisonerID→Prisoners, CrimeID→Crime, CrimeDate)
Visits          (VisitID, PrisonerID→Prisoners, VisitorID→Visitors, VisitDate)
EmergencyBackup (EmergencyBackupID, EmergencyID→Emergencies, BackupID→BackupForces)
Users           (UserID, Username[unique], Password, Role, StaffID→Staff, GuardID→Guards)
```
